# Collective Trauma Summit 2022

CREATING A GLOBAL HEALING MOVEMENT

## *Creating Environments of Safety and Trust Through Polyvagal Theory*


FEATURED SPEAKER

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HOSTED BY

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Thomas Hubl: Hello and welcome back to the *Collective Trauma Summit*. My name is Thomas Hubl, I'm the convener of the Summit, and I have the deep honor and pleasure to sit here with Dr. Steve Porges again. So welcome, Steve, to our Summit again.

Stephen Porges: Well, thank you, Thomas. It's a pleasure to see you again and looking forward to an interesting dialogue.

Thomas: Yes, me too. You know, often when I travel, when I give talks, I often refer to your work because I think it's such a profound contribution to any kind of trauma healing work and any kind of also difficult crisis environments where we can take so much from your work and where you contributed to the world. So, first of all, thank you for that. I think it's a real gem and it helps us to bring the healing of trauma closer to many people that are not necessarily specialists in the first place, so that's really, really great. So thank you for that.

Stephen: Thanks for the kind words.

Thomas: First of all, I'm interested—you know, we have had already multiple conversations here throughout the years, and of course I'm always interested—you spend so much time first researching then teaching and it's spread around the world, but what's your leading edge at the moment? What excites you right now? Where do you see is the leading edge of the polyvagal theory work in the world?

Stephen: Well, it's an interesting—and I would say almost a profound—question, because the theory kind of emerged from basic research and it was really getting the landscape of what our nervous system is and what we have and where it came from. So our evolutionary history became very important in understanding the resources that we bring into the world we live in.

I would say that the most important thing that's happened to me over the journey of, after coming up with the initial theory, was really the understanding of trauma—the human experience from the world of trauma—because it basically was following the roadmap of the theory. It was basically doing de-evolution or dissolution evolution in reverse.

As we are challenged to survive we go down to our older, more foundational survival mechanisms, and it was very insightful to see this occurring, because what it taught me—literally from the lens of a traumatized individual or survivor of severe trauma—is what is missing, what is lost from trauma. They never gave up that dream. So the individuals who have survived trauma still have this visualization, this dream of feeling safe in the arms of another.

It is a very interesting thing to talk about what's innately in our body and to respect it and to understand how we can, in a sense, convince our body that it can be safe enough to be in proximity with others. That we can trust, we can love. It's been a very interesting journey in terms of reframing my own perspective of the world, which like many academics is very, I would say, intentionally oriented, where top down thoughts …

So it's like you see the world with basically ethical principles and you want it to behave that way and you don't understand why it isn't until you get into the world of trauma. Then you see that the physiology has … it rules. The physiology rules, not the intention. So we can understand the aggressiveness that people exhibit, and the word 'defensive'. When we're defensive, it's not a volitional, intentional defensiveness. We are in a state of defense.

Thomas: Right. That's beautiful. Maybe you can deepen a little bit for all the new listeners that didn't hear your talks before. Maybe you can deepen a little bit what happens actually in the nervous system when overwhelming situations happen.

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Stephen: Well, I was going to fold it into the concept of what's gone on during the pandemic, and that is that ... Because a defense is triggered not only by psychological threats, but by pathogens. So we have this conceptualization that if it's not physiological, meaning it's not viral, then we should be able to control it. We should have the intention of not. We don't understand, or I would say acknowledge, that our nervous system doesn't really care whether it's a physical threat or a psychological one.

The nervous system is experiencing threat. Now, what does that do? It shifts the non regulation and now we can get into a theoretical perspective. The polyvagal theory really says that in between the experience there are our behavior and the context in which our behavior occurs, so you can call it cause and effect or stimulus and response, our organism resides.

So the problem has been that we have lived in a scientific world that really rewards and wants cause and effect relationships as laws of science, and the behavioristic world which has stimulus drive responses. We have neglected—and abusively neglected—respecting the body. The body is in between. And what is the body? It's the nervous system in polyvagal theory.

That nervous system is regulating autonomic states—the state that our organs in our body are—so that when we're under fear, we can mobilize. It helps our heart rate beat fast. It gets our muscles stronger, we get aggressive. But when we're overwhelmed, it also has a metabolically conservative response that shuts us down. Of course, people in the world of trauma or people in the world of overwhelming stress know what I'm talking about. The body just can't go anymore.

So what happens is that we have this need to get the respect for the physiological state. In polyvagal theory, if there's one thing to teach people, it is that our body is important—it's not just our mind, our intentions—and we can become aware of our body and we can respect what our body is screaming at us. Of course, our society—and this is the extended Western society—says, "Don't listen to the body. Keep moving, keep working."

Of course, there's a great price to pay for that, and we see this in terms of what used to be called psychosomatic illnesses and diseases that now carry terms like functional disorders, because they can't find measurements yet in the end organs. But really it's saying the nervous system has given up. It's no longer regulating your organs, and over time those organs are going to become diseased.

So the problem is that the measurement of nervous system regulation within medicine is primitive at best. In fact, it's neglected. When you go to a specialist, they're really looking at tests of the tissue or markers in the bloodstream. They're not

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looking at how these organs are following feedback loops. The feedback loops are listening to the organs and telling the organs what to do. Psychologically, when we numb out and tell our body to do things that we don't want to do, we are impeding our body's natural feedback loops.

Thomas: That's very powerful. So that also says in a way that there is a whole sophisticated information technology in the body, and there are many, many feedback loops. When they break down, then we naturally see diseases as a result of it. So it links trauma directly to physical—

Stephen: Yeah. Think for a moment about how people who have survived trauma have been treated. They've been treated, say, "Oh, you weren't physically injured." Hey, now let's get realistic. They were physically injured. The nervous system went through an experience of a threat of life. So there's really acknowledgement. I want to bring up something that has really been ...

You asked me, what am I thinking about? What is the passion? So I spent, I would say, the last few years thinking about evolution leading to sociality, and sociality leading to co-regulation. And our heritage is that we take care of each other, we calm down. We know that—looking at the mother and the baby—that becomes our model.

When the baby is losing regulatory state, the mother's prosodic intonation of voice, gestures, and engagement calms the baby. The mother's not yelling at the baby and saying, "Stop crying." Although you may find instances of that, it doesn't work. So our nervous systems know what cues of safety are. When you have pets, you talk to your cat, dog, horse, mammals, social mammals.

We all share this need for intonation of voice as a signal that we are safe. But what I've been working on now, so I actually said, "Okay, that's an interesting journey," but you know what it does? Number one, it elevates sociology as a neuromodulate, which is wonderful. So we now have a portal of treatment, and every intuitive therapist gets that. Every intuitive parent gets that.

Every loving person gets it that our connectedness with others is a portal towards healing. So we get that part, but from evolutionary theory, we always start asking questions. First of all—I'm not an evolutionary theorist, fortunately, so I can make certain types of statements—but you have parallel evolution, meaning within the human or social mammal you have this expansion of cortex; you have a bigger brain and that bigger brain needs lots and lots of oxygen.

I've always saw this as sociality as the product, and this is just wonderful, but let's

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move it to a more primitive, or let's say basic level. As the nervous system, or as the social mammal starts getting this big mass of neural tissue on top of the brain stem—our neocortex—it was metabolically very, very costly to keep oxygenated blood to that area.

So what it meant was that the traditional—and this is kind of a slippery word, which I really mean by ancient—the traditional defenses of vertebrates were fight/flight or shut down. Now, the problem is that mammals, since they have such a great need for oxygen, they could not fight or flee for long periods of time and shutting down was metabolically destructive.

The evolutionary journey as a tinkerer, the evolution as being a tinkerer or working with existing structures, not designing, not starting from the beginning, but tinkering. You have this reptilian brain and you become as mammal. What are the changes? The changes are the cortex, the changes are greater oxygen needs, the changes are sociality, which we've always focused on.

But practically what that body needs is the ability not to be in the state of defense so it can heal. That is really what sociality is all about. It's about sending the appropriate cues to turn off hyper vigilance/defensiveness to enable oxygenated blood. Now, the ancient vertebra could calm down, they just shut down totally. In doing that they reduced oxygen, so that mechanism was deadly. It was lethal.

So you have this mammalian immobilization, which you see with the mother and the baby. So you have this literally hands shaking of a nervous system, or a system that requires oxygen, and a big brain that requires oxygen. Then you have sociality, which enables oxygen to get to the brain. So it's like twisting a Rubik's cube, and if you are a humanist, you see sociality, benevolence, and compassion.

If you're, in a sense—I was going to use the word—an engineer, and you're looking at oxygen utilization and resources, you start seeing, "Can I slow the activity of the organism while giving it oxygen?" You start looking at different pragmatic aspects. So I see it as literally a beautiful story of how our ability to connect and interact and to love is actually deeply rooted in the need, or in say the process, of enabling our tissues to heal. So it's a beautiful statement in a sense, say, the feelings of safety enable us to love, to care and helps our organs as well.

Thomas: Wow. That was a beautiful framing of how you brought all these parts together. I really love it. It's great. It's a lovely arc that you introduced many, many important things together. It's beautiful. So when we look at that now, and we say okay ... Because in my experience, we do a lot of work in large groups and the synchronization of large groups.

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I think especially the polyvagal theory, the co-regulation, and the I feel you feeling me is a great way online and offline in groups spaces to create intimacy. Maybe you can speak a little bit to how we can create healing environments or social environments that do exactly what you described before in groups and why groups are so powerful.

Stephen: You're basically triggering some earlier thoughts that I've had for several decades. I was very interested in designing classrooms. Actually, I was interested in designing healing environments, and that's really what you're asking. I looked at neonatal intensive care units, I looked at classrooms, I looked at workspaces, and the objective of the design of those spaces is surveillance and cleanliness. It has nothing to do with physiology.

So the sensory experience of those spaces are really triggers of defense, not triggers of openness and curiosity and learning and caring. So what you're bringing up is first the actual environmental cues, the physical cues in which groups are set or a therapist has their clinic. The physical environment is critical because the background sounds have to be minimal. There can't be mechanical rumbles because our nervous system detects low frequency sounds as predator and it detects high frequency sounds as danger.

So it is a band of frequencies primarily defined by a mother's prosody, or a male talking to their pets. When we talk to our pets, we raise the tone a little bit because we intuitively know how to calm another social mammal. So the first part is getting rid of or minimizing the distractors—that in polyvagal terms—result in a neuroception of threat. Again, what is polyvagal theory all about? It's a science of safety.

It's all about understanding the innate needs, the quest to feel safe. If we translate feeling safe to saying, "I feel safe when my autonomic nervous system is doing its job. It's supporting health growth and restoration. When I don't feel safe, my autonomic nervous system is supporting defense." We can see this in people who suffer from anxiety disorders or developmental disabilities like autism.

We can see this occurring. So the first thing is the physical context. Don't make it too hard for the nervous system. In terms of designing space, you can use sound absorbent wall hangings or floors. People tend to forget about the importance of echo or how sound bounces off hard surfaces. So that's number one. The other one is visual distractors.

When I designed—or helped design—a school on autism, we raised the windows so

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that they didn't look at the street, they looked at the trees. So you got lots of ambient light coming through, but there were no distractors. People felt they were in a forest. It was really very nice. So this is all the prerequisites.

The prerequisites are really the physical environment. Don't challenge the nervous system too much. The next part is really how are the interactions working, or why do they help? The issue is, if I'm not distracted, my body can look at you and if you look back at me ... We use terms like mirroring, but really what we're doing is we're sending cues that our nervous system detects as cues of safety; facial expressivity, intonation of voice.

Familiarity is important, because you run groups and what you notice in groups is how the, I would say, the texture or cohesiveness of the group changes over time because they become familiar with each other, and our nervous system interprets familiarity as safety. So even though there's uniquenesses among the people, it's who they are, and we become, as a species, quite accepting.

Thomas: Yeah, that's beautiful. So in my experience, the groups develop different levels of coherence with each other, different levels of safety, which allows the nervous system also to detox different levels of its own trauma storage. So there's a whole digestion process that reinforces, again, resilience or grounding or openness. So we can see that in a regular group setting that happens in a safe space.

Now, a lot of the collective trauma work, unfortunately, is happening also in crisis area like Ukraine, for example, now, so we are running a course for NGOs; trauma informed leadership in NGOs in crisis areas. Maybe you can also speak a little, and I'm sure you also have some experience, how we apply the findings of the polyvagal theory in hot crisis areas where people are really under danger, where it's not safe. What's the best application for people that live in that kind of circumstance?

Stephen: Yeah. I've been online with therapists in Ukraine twice and once in Poland and it's very interesting, challenging to kind of experience the world they're in at that moment. The experiences that they are sharing with me are ... I guess there's certain rules that I can extract from the dialogues I've had with them, and that is you can't be a good co-regulator of anyone if your body's in a state of threat.

I mean, it just starts right there. So the first rule is an awareness of where you are and also understanding that this is challenging, because the threats are real. But the job of the therapist is really to create the context, the social context, to co-regulate with the client. They had a very interesting experience in Poland. They wanted to utilize my safe and sound protocol, which is this acoustic intervention.

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Actually, I said, "Probably not." Because you first had to be safe, and what they wanted to do was work with parents to work with children. The first rule was the therapist has to be aware of themselves and safe, the parent has to be aware of themselves and safe, before they could even work with the child. But the interesting thing was one of the therapists had tried the safe and sound protocol, which really is ... I view it as a stealth intervention, because it's broadcasting cues of safety.

The theory underneath it was that when our nervous system detects cues of safety, the social engagement system—meaning the muscles of the face and head and the vagal regulation of the heart—move into a state of engagement co-regulation and, in a sense, more powerful vagal regulation to support homeostasis. So what happened? The first person they tried was a woman and she lasted 45 seconds.

What happened was the cues of safety come in, and they come in and so a person who's escaping threat, and now you give them this sense of safety. Their body should be accessible, but now as the body opens up it's interpreted through enteroception, meaning that my bodily feelings and my bodily state are now being interpreted by my conscious mind.

And that results in, "Got to get out of here," because it's vulnerability, and it's really remarkably reliable. It was a very important lesson for the therapist to learn that first they have to be safe. Second, if they want to work with children, the parents have to be safe. Safety is not a pill. It's not a manipulation. It's the body accepting those around you. So if you want to think about what is safety.

When I feel safe, I feel that others around me are welcome. If you come out of a war zone, your body is retuned to be skeptical, functionally skeptical. We use terms like hyper vigilance. But what it is saying is my nervous system has to be reassured that everything around me is safe. Now, intellectually, I may know that, but when you're in situations or have experience, bombing situations, the nervous system gets very much retuned.

So I was also online, like I said, twice with a therapist in Ukraine, and it's really remarkable the optimism and the spirit of people. So they came out to hear me in a talk—I was talking to ... In a Q&A—and they came out of a bomb shelter. So my first question to them was what did they do in the bomb shelter? Now, this gets at the root of what we're going to all talk about. They said, "Well, we were singing."

I said, "What were you singing about?" I was basically explaining that they were recruiting the social engagement system through vocalizations, through song to make their bodies feel safe and to co-regulate each other. It became a very important point for them to understand because they were, in a sense, co-regulating, which

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was wonderful. They came out smiling.

Coming out of a bomb shelter is not ... And being interactive on a webinar, I felt that was just totally remarkable. But the issue is intuitively they knew what to do, that they were, by singing ... It's slow exhalation, it's utilizing the muscle of the face and head, and that stimulates in the brain stem this area called the ventral vagal complex.

That is soothing. It calms the heart rate. So listening to singing stimulates basically sensors on the eardrum itself. So when the eardrum is moving and tensing, it's sending signals in as well. So vocalizations, singing, and even listening to that are techniques of getting our bodies in calmer states.

Thomas: Yeah. That's beautiful. Yeah, I also experienced this. I think it was last year or the year before there were missile attacks on Tel Aviv and a lot of sirens go off and it's a little bit of a similar experience that you shared, like how I regulated myself and then from that I could be with my daughter, and it's a scary situation for kids. Then how we create a social environment within a "dangerous" situation. So that's a pretty much along the lines that you said. We also experienced that.

Stephen: Intuitively we reach into top down visualizations of positive things, which are really affecting brain stem areas. So if you have a visualization of a loved one and a wonderful moment, it's difficult to be scared or to feel bad because it's a powerful input to that same area of the brain stem. But if you're interacting with others—singing, talking, smiling, telling jokes, but in close proximity—it's very powerful and it's been really part of our basic toolkit since the evolution of humans. Actually, evolution of social mammals, other social mammals who will gather and co-regulate each other.

Thomas: Yeah, that's right. What would you suggest in the areas where, for example, many people are triggered and there aren't many people that ... Where do we start when a community is very activated or very stressed? So where would we start to bring in resources into such areas? Do you have some experience or did you have some thoughts about that?

Stephen: Well, I don't have experience and I think ... Okay, there's interesting sort of issues going on. We have war, we have pandemic, we have economic crisis, and in the United States we have political discord, but in Israel you have something similar going on. So the issue is what is it that we really want to do? We want to feel safe with others.

In a way, we're less concerned about political events or things like that than we are

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about feeling safe in the presence of others. It's a very interesting feeling, and what does that mean? It means autonomically they move into a defensive state, which functionally means they're less accepting of others. So it doesn't matter what your position is.

If you are angry, mobilized, or frightened, you're not accepting of others. That's not the way our species evolved. That's not the way we thrive. We thrive by basically being accessible, not frightening each other. So you ask the question, how would I do this? Well, the issue is first, I guess, we have to acknowledge what the real threats are. I mean, this becomes ...

And then we have to understand how those threats affect our physiology, how we feel. Then if we are—and we think we are—problem solvers as a species, we know that we can't solve problems, at least efficiently and effectively, when we're in a great threat. So we know on our own individual level that our brains and our problem solving capacities are dependent upon feeling safe enough.

So is our creativity and so is our generosity. Feeling safe enough, by the way, makes us into this wonderful—I wouldn't say makes us—enables us because we are a benevolent species, but under states of threat, we aren't. I think we have to understand that, that as we scare each other—and it's reciprocal—we lose the wonderful capacity of our own humanity, and that is the ability to be accessible and to care for others.

Thomas: That's very true. One thing I would love to hear your take on is ... Because what we see in the trauma work often ... I call this the individual, ancestral, and collective trauma work, as one system. So when we see there's a transgenerational transmission, we see it here, for example, in Israel, the transmission of Holocaust survivors, first, second, third generation, or of course in other places.

So when we look at the polyvagal theory and co-regulation and let's say the epigenetic changes that happen through trauma, how do you see how might our way ... So if somebody does the inner work, we become better co-regulators, we regulate ourselves more, what is the rebound effect, maybe, on the epigenetic map that we have been born with?

Stephen: Okay. Polyvagal theory is optimistic. It's an optimistic theory because it says that much of our behavior and many of our feelings are not genetically determined. They're not even epigenetically modified. We just get locked into different states and those states are relatively malleable, flexible. We have to understand that our body evolved with a mechanism to turn off our defenses.

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That was through social engagement, through trust. We use words like love, but we all know what we mean. It's not love in the same passion. It's love as trust, that we know that we will not be harmed by another. That is really, I think, missing so much in our society. We are cautious that we will be harmed. So the bit is ...

I've pondered on this issue of transgenerational trauma and I've thrown out the term intergenerational, transgenerational, because I think it's a long history.It's a history of humanity. And I've come up with terms or words, like we are a traumatized species and trying to understand that and understand what it means and what does it enable us. What's our flexibility?

I think that's your question. What's our flexibility? I think as we start talking about genes and methylated genes during epigenetic transitions, we lose this flexibility of thought, perhaps. But let's take the optimistic view and say, "Look, I can be in a state of anger and frustration, and I'm not a very nice guy. But I can take a couple breaths, I can walk on the beach, I can see people, I can smile, and I can become accessible."

I think also like in many therapeutic models there's parts work they talk about, whether it's internal family systems or other models, and I think if we modified it and not thought of these parts as totally self-organized and self-protecting but saw them as derivative of being in a different physiological state, then we would be, in a way, as we respect the parts, we're respecting our physiological state and we're learning how to choreograph and organize that.

We can see that there's this social person, the social engagement system who's totally regulated, in control and aware of all. Then you have a defensive state which is aggressive, and you have a defensive state that's withdrawn. Many of the parts clearly map into this, but what you have is you have many more parts than some people and that's because you start moving up higher and higher in the brain and the brain will just start creating narratives.

But you have three primary plots, primary narratives, and they're survival narratives. They are our genetic history. They are our heritage and that is really behind the therapeutic interventions is the repurposing of parts, but it really follows the evolutionary tradition. So what does evolution in repurposing structures? We repurposed our shutting down response into being able to immobilize without fear, to be intimate, and to optimize our healing.

We have repurposed our fight/flight mobilization system. We call it play, exuberance, dancing, moving, team sports, and we are able to, in a sense, repurpose these because we never turn off our social engagement system, it's always there. So we don't say, "Leave me alone so I can optimize my energy to be a fight and flee." We

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say, "Come with me, watch what I'm doing. Smile. Isn't it wonderful what I can do and how I can interact with you?"

So we are carrying our self-awareness and our ability to connect wherever we go within our own physiological history. I think that's what therapy is all about. I think our journey in life is about respecting, honoring what we come into the world with, these different systems, and utilizing that both for our own personal pleasure and experience, but also for the betterment of humanity.

Thomas: Right. Beautiful. When you look at, let's say, the societal fabric, and you see a lot of polarization at the moment—and in this Summit we have a special track on interviewing some people that come from journalism and media—and we look at, okay, what's trauma informed media, what's polyvagaly informed media in-formation? Maybe you can speak a little bit to both sides, how media can respect the principles of the polyvagal theory, but also how media consumption. It's a two-way flow.

Stephen: Yeah. As you talk about, you also go into the notion of addiction or media addiction, and addictive behaviors are effective and efficient methods of shifting our physiological state, but not necessarily into a state that is accessible and benevolent and co-regulatory. I think in journalism, a journalist actually has to start taking some responsibility of literally being fear-mongering, because the impact of ...

Okay, so the conspiracy theorists, which are big in social media and have actually infiltrated the American politics, are all about fear. It's all about exploitation. It's all about loss of voice. It's very predictable about how to get people upset, you take their voice away. That was like fraud in election, take your voice away and you don't exist. No one's listening to you.

Rather than being told you're stupid or [*inaudible*] these things, a different strategy might be to educate or to be welcoming and, to a sense, honor the fear, but not honor the so-called lack of evidence that supports the fear so that people can be trusting of a portal or conduit of information. I think if we go back to before social media and we start talking about network news of the 1960s, there was a better, let's say a more ...

Basically people believed what they were hearing, and now everyone's skeptical and the news channels are extremely polarizing. I would say during the January 6th hearings, which are going on, I would say the best part of the television coverage has been that there's no commentary.

You're basically listening to people, and what you hear are people with beliefs, with

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ethics, with passion, and you may not agree with their political perspective, but you believe that they are trying to do their best. I think that is very important for us to understand is that we may not agree with everyone, or we may disagree passionately, but we can be respectful of people when they are seeking truth.

Thomas: Yeah, that's true. Very true.

Stephen: Think of it as trauma therapy. When the therapist is supportive and witnessing, therapeutic presence, what's it doing to the client? It's enabling the client to express. But if the therapist intrudes and starts to reinterpret without even having the client express themselves, that disruption creates a feeling that they're not being heard. I think the commission's public hearings of the January 6th commission enabled a type of resolution for those who were talking and for those of us who are listening. Our bodies, like in trauma, need to resolve the jolts.

Thomas: So when we take that now and we say, in your overall work ... So I think polyvagal theory really spread very far, it inspired many people, including myself, and I think it's being part of many modalities of trauma therapy, and also the leading edge modalities of trauma therapy I think are very much based on co-regulation. So when you see your own work expand into the world, are there specific projects that you are very passionate about, how you deepen the understanding of polyvagal theory in the world?

Stephen: Yeah. I'm very committed to changing medical education. Okay. So the trauma world has been welcoming and polyvagal theory kind of fits nicely into it, but there's still a basic disconnect when people go to see their physicians. The interesting part of what we're learning, or I'm learning, is that there's so many clinical symptoms that cannot be addressed with medicine. So in a sense, they're called functional.

It used to be called medically unexplained symptoms, but they're all very predictable by polyvagal theory. They're basically saying the body's moving into a state of defense. So I've actually created a core of academic physicians. I figured if we're going to do something they'd better be in the academy, because you don't want outsiders telling insiders what to do.

So we have a collection, a group of about six now, academic physicians, trying to develop course material for physicians. It's been a very interesting set of meetings because the question is, do you change medical school curriculum or do you go in and deal with continuing education? Because physicians are frustrated, they're angry because they're not getting co-regulatory experiences.

The bottom line is that they have been ... In modern medicine, they're not, in a sense,

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allowed in many situations to spend sufficient time talking to their patients. So they're not getting co-regulated and they're, in a sense, burning out. They're being a dispenser of information, of information that is not being accessed by the client because they're in a state of fear and everything is falling apart.

There was an anesthesiologist who's quite remarkable. She gave grand rounds on polyvagal theory in a major medical school and basically everyone there loved it and they started asking her to give grand rounds, lectures, to other departments because what she was talking about was how, even in anesthesia, how you work with your clients, your patients, enables whether they can literally tolerate the operating room experience.

So in a sense, enabling them to feel safe enough, and she's just remarkable. She has this very calm way of interacting. So there are people out there that we're trying to get together so that we can start creating materials. The other one that I'm, I would say, very passionate about is education. That we have situations in education where children are literally being traumatized by being, in a sense, going to school, being bullied.

They're telling you, their bodies are telling you. There's tremendous gut problems, cyclic vomiting, nausea, and these are really—from polyvagal theory—symptoms of life threat. So the body's telling you. What do the parents and what does the school say? "Get over it, go to school." Fortunately, I'm getting involved in research with these populations and we can see in these populations a dysregulation of the autonomic nervous system.

So we're finding very powerful indices that the nervous system has already been retuned to be defensive. I have a metric that I've developed called vagal efficiency and basically it becomes an objective measure of many of the ... Detects many of the individuals that we've received a diagnosis of dysautonomia. This is really quite remarkable.

Now, with that measure and with these populations, we can start seeing if interventions change the neural organization. So that's where I'm moving on that. COVID also created another platform for all these things, because COVID—like any other chronic disease or illness—is really the nervous system has effectively fought off the pathogen or the damage, but the nervous system functionally didn't get the message.

So the symptoms remain, but the disease has been ... The nervous system won the battle, but didn't tell its defense systems to stand down. It's kind of an interesting model and that, I think, can be really ... I think polyvagal theory can be very useful by

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giving cues of safety to that nervous system so it can literally calm down again. There are several therapists working with long COVID who are polyvagal informed, and many are using the safe and sound protocol and they're getting results.

But the issue is how do you convince the nervous system that it's no longer under a state of threat? This is the trauma world, but now you go into something like chronic fatigue syndrome, Lyme disease, and long COVID. These are similar features. The nervous system didn't get the message: it's okay to be accessible.

Thomas: It's powerful also what you said, that through applications of polyvagal theory can literally affect long COVID. I mean, that's in itself a very strong message, a very hopeful message.

Stephen: Yeah. It's a theory of optimism, because it's saying our nervous system is smart, but it's smarter to deal with disease and so it makes a conservative decision. It gets into a defense mode. It's not very easy to get that out. It's not necessarily epigenetic, it's not hardwired. It just has to be convinced. How do you convince it?

Polyvagal theory is identifying portals through which you can convince the nervous system to slowly open the gates. Not fast. So it's like the woman from Ukraine who was being treated in Poland for basically PTSD symptoms, and the initial listening to the music she becomes like this, and almost beatific, and suddenly it's like this and she's out of there, because it's vulnerability.

So we start understanding this, but it doesn't mean that she won't later be able to go very slowly through listening to these sounds and experiencing it for short periods of time to convince her nervous system that short periods of time are safe enough to be ... It's okay to be vulnerable for short periods, and then we can start expanding it. That's, in a way, what therapy does as well. So there are all these principles, but in the polyvagal world there's a physiological substrate that is going along with the emotional and behavioral features, that clinicians are safe.

Thomas: Yeah. That's beautiful also. We are also working in Spain with a clinic, a pediatric clinic, and maybe it's going to expand into the area on the same principles, and it's so welcome because it's so needed to have co-regulation. Also between the medical stuff and support systems in hospitals, this is so important. First of all, I really enjoy listening to you. I think you have a beautiful way of framing all of that, which is on the one hand complex, but it makes it very simple to understand. If you look at ...

Maybe you can leave us with ... Because for many people looking at our world today, we see a lot of polarization, we see a lot of ... More and more climate events are coming towards our ways. Most probably many millions of climate refugees will

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come, sooner or later. So for some people it feels like, "Wow, there are so many impacts," and media and social media, it brings us a lot of trauma impulses.

A school shooting here, or something there. I believe what you said, there's a very optimistic theory that we can really do something to find our centeredness, our resilience within all of that. Maybe you can speak a little bit to how some people might interpret our time as a bit dark and difficult to deal with, and then we hear the optimism of polyvagal theory.

Stephen: First of all, let's acknowledge that there are threats to us in our society. So the first thing is to realize that if we are to solve severe problems of our survival as a species, we have to be safe enough with others to enable our problem solving capacities to be expressed. That if we are frightened and scared, we cannot contribute. That's our paradox.

So the important thing is we need to be safe enough for a sufficient amount of time every day—or every other day, or something—to enable our creativity as well as our benevolence to occur. If we become totally frightened our skillset gets limited and, in a sense, there's a pragmatism here. Coupled with the optimism is a pragmatism, and that is that if your body starts becoming too frightened, then you can't solve the problem.

Now, that is a problem in itself. So we're smart as a species. It means we need social relationships. We need co-regulation. We need interactions. In the world of therapy, it means that therapists need to basically be self-compassionate. They need self care. They need to respect who they are. I think it's a general theme across all of our professions and disciplines is that we can't be benevolent and compassionate if we're in the state of threat, and we can't even be healthy if we're in chronic states of threat.

So the issue is we need to know that threat is a mortal enemy to our potential attributes. That's a problem that we need to focus on. In a funny way, our educational models never focused on that, they focused on more information, more cognition. They didn't focus on a basis of sociality, of co-regulation, of friendships and trust.

Now, I want you to sit back for a moment and visualize an educational model that's based upon a foundation of sociality and trust. Think about a society based upon sociality and trust, and even think about a business world. Because everything that enables us to function as, let's say, as human beings—our health, our interactions—require our bodies not to be in states of threat.

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This is not necessarily what we've been taught. We've been taught not to listen to those bodily needs, to produce, to work harder, basically to run faster, and we've also been taught—in many situations—not to trust others. So we've been taught all the wrong things about what our nervous system evolved to do. So anyway, Thomas, I welcome you into my world.

Thomas: Thank you, Steve. It's such a delight to be with you, and even there's your whole knowledge and your academic background, but then there's the beauty of your being and you transmit really what you are speaking about, so that's really very pleasant. It was a great arc that we went through, and I think many of our listeners will get a lot out of this contribution of yours to the Summit. So thank you very much, and I'm very happy for our unfolding collaboration. So carrying your work into the world all over, wherever I go, so I'm very grateful to you and your work.

Stephen: Well, I look forward to the day where we can meet face to face.

Thomas: Yeah, me too. Me too, Steve. So thank you very much.

Stephen: Thank you, Thomas.

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